PCS-302 : Data Structures Lab

Sorted Matrix Search | O(n) Linear-Time Search in a Row- and Column-Sorted Matrix

Problem Statement

Given a matrix of size n × n, where every row and every column is sorted in increasing order, design an algorithm and write a program to determine whether a given key element is present in the matrix. The solution must run in linear time, O(n).

Example

```
Matrix (4x4, rows & columns sorted increasing):

   10  20  30  40
   15  25  35  45
   27  29  37  48
   32  33  39  50

Key = 29  -->  Found at row 2, col 1
Key = 100 -->  Not Found
```

Algorithm (Staircase Search)

```
1. Start at the top-right corner: row = 0, col = n - 1

2. While row < n AND col >= 0:
     a. If mat[row][col] == key:
            return (row, col)              // element found
     b. Else if mat[row][col] > key:
            col = col - 1                  // move one column left
     c. Else:
            row = row + 1                  // move one row down

3. If the loop ends without a match, return NOT FOUND

Time Complexity : O(n)  -- each comparison eliminates
                           either one full row or one full column.
Space Complexity: O(1)
```

C Program

```
#include <stdio.h>
#define N 4

int main() {
    int mat[N][N] = {
        {10, 20, 30, 40},
        {15, 25, 35, 45},
        {27, 29, 37, 48},
        {32, 33, 39, 50}
    };

    int key, row, col;
```

```
    int found = 0;   // 0 = not found, 1 = found

    printf("Enter key to search: ");
    scanf("%d", &key);

    // Start at the top-right corner
    for (row = 0, col = N - 1; row < N && col >= 0; ) {
        if (mat[row][col] == key) {
            found = 1;
            break;              // key found, stop searching
        }
        else if (mat[row][col] > key) {
            col--;              // too big -> move left
        }
        else {
            row++;              // too small -> move down
        }
    }

    if (found == 1)
        printf("Element %d found at position (%d, %d)\n", key, row, col);
    else
        printf("Element %d not found in the matrix\n", key);

    return 0;
}
```